

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

# ***B.Tech. Degree IV Semester Examination April 2019***

## **ME 15-1405 HYDRAULIC MACHINERY (2015 Scheme)**

Time: 3 Hours

Maximum Marks: 60

### **PART A** (Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Find the force exerted by a jet of water of diameter 100 mm on a stationary flat plate, when the jet strikes the plate normally with a velocity of 30 m/s.
- (b) What do you mean by repeating variables? How are the repeating variables selected for dimensional analysis?
- (c) What is meant by geometric, kinematic and dynamic similarities?
- (d) Define and explain hydraulic efficiency, mechanical efficiency and overall efficiency of a turbine.
- (e) What is cavitation? How can it be avoided in reaction turbines?
- (f) Explain the working of a single-stage centrifugal pump with a neat sketch.
- (g) Define slip, percentage slip and negative slip of a reciprocating pump.
- (h) Draw an indicator diagram, considering the effect of acceleration and friction in suction and delivery pipes of a reciprocating pump.
- (i) A hydraulic press has a ram of 300 mm diameter and a plunger of 45 mm diameter. Find the weight lifted by the hydraulic press when the force applied at the plunger is 60 N.
- (j) Differentiate between forced vortex and free vortex.

### **PART B**

(4 × 10 = 40)

- II. (a) State Buckingham's  $\pi$ -theorem. (2)
- (b) The force exerted by a flowing fluid on a stationary body depends upon the length (L) of the body, velocity (V) of the fluid, density ( $\rho$ ) of fluid, viscosity ( $\mu$ ) of the fluid and acceleration (g) due to gravity. Find an expression for the force using dimensional analysis. (8)

### **OR**

- III. (a) A jet of water of diameter 7.5 cm strikes a curved plate at its centre with a velocity of 20 m/s. The curved plate is moving with a velocity of 8 m/s in the direction of the jet. The jet is deflected through an angle of  $165^\circ$ . Assuming the plate to be smooth, determine: (5)
  - (i) Force exerted on the plate in the direction of jet
  - (ii) Power of the jet
  - (iii) Efficiency of the jet
- (b) A pipe of diameter 1.8 m is required to transport an oil of specific gravity 0.8 and viscosity 0.04 poise at the rate of  $4 \text{ m}^3/\text{s}$ . Tests were conducted on a 20 cm diameter pipe using water at  $20^\circ\text{C}$ . Find the velocity and rate of flow in the model. Viscosity of water at  $20^\circ\text{C} = 0.01 \text{ poise}$ . (5)

(P.T.O.)

- IV. (a) A Pelton wheel has a mean bucket speed of 10 metres per second with a jet of water flowing at the rate of 700 litres/second under a head of 40 metres. The buckets deflect the jet through an angle of  $160^\circ$ . Calculate the power given by water to the runner and the hydraulic efficiency of the turbine. Assume coefficient of velocity as 0.97. (5)
- (b) A Kaplan turbine working under a head of 20 m develops 12000 kW shaft power. The outer diameter of the runner is 3.5 m and hub diameter is 1.8 m. The guide blade angle at the extreme edge of the runner is  $35^\circ$ . The hydraulic and overall efficiencies of the turbine are 88% and 84% respectively. If the velocity of whirl is zero at outlet, determine: (5)
- Runner vane angles at inlet and outlet at the extreme edge of the runner.
  - Speed of the turbine

OR

- V. (a) What is meant by the speed ratio of a Pelton wheel? (2)
- (b) What is a draft-tube? What are its functions? (3)
- (c) A Francis turbine with an overall efficiency of 75% is required to produce 149 kW power. It is working under a head of 8 m. The peripheral velocity  $= 0.26\sqrt{2gH}$  and the radial velocity of flow at inlet is  $0.96\sqrt{2gH}$ . The wheel runs at 150 rpm and the hydraulic losses in the turbine are 22% of the available energy. Assuming radial discharge, determine: (5)
- The guide blade angle
  - The wheel vane angle at inlet
  - Diameter of the wheel at inlet
  - Width of the wheel at inlet

- VI. (a) A centrifugal pump is to discharge  $0.12 \text{ m}^3/\text{s}$  at a speed of 1500 rpm against a head of 25 m. The impeller diameter is 250 mm, its width at outlet is 50 mm and manometric efficiency is 75%. Determine the vane angle at the outer periphery of the impeller. (5)
- (b) Find the number of pumps required to take water from a deep well under a total head of 90 m. All the pumps are identical and are running at 800 rpm. The specific speed of each pump is given as 25 while the rated capacity of each pump is  $0.16 \text{ m}^3/\text{s}$ . (5)

OR

- VII. (a) A single-acting reciprocating pump, running at 50 rpm, delivers  $0.01 \text{ m}^3/\text{s}$  of water. The diameter of the piston is 200 mm and stroke length 400 mm. Determine: (i) the theoretical discharge of the pump (ii) co-efficient of discharge (iii) slip and the percentage slip of the pump. (5)
- (b) The length and diameter of a suction pipe of a single-acting reciprocating pump are 5 m and 10 cm respectively. The pump has a plunger of diameter 15 cm and a stroke length of 35 cm. The centre of the pump is 3 m above the water surface in the pump. The atmospheric pressure head is 10.3 m of water and pump is running at 35 rpm. Determine: (5)
- Pressure head due to acceleration at the beginning of the suction stroke
  - Maximum pressure head due to acceleration
  - Pressure head in the cylinder at the beginning and at the end of the stroke

- VIII. Explain with the help of a neat sketch, the principle and working of the following hydraulic devices: (10)
- Hydraulic intensifier
  - Hydraulic crane

OR

- IX. Explain with neat sketches, the working of hydraulic torque converter and gear pump. (10)